

- 6 (a) (i) Define *decay constant*.

The **decay constant**, λ , is the probability of decay per unit time of a nucleus

.....[1]

- (ii) Suggest why the determination of decay constant by measuring the mass and activity of a sample can be used only for nuclides that have relatively *small* decay constant.

If the decay constant is small, the activity will be small and thus there will be ample data to be collected. The data sets need to be large so that an accurate value of probability can be deduced for any random event.

.....[1]

- (iii) Explain why the random nature of radioactive decay makes it difficult to measure the decay constant to a high degree of accuracy.

To minimise the effects of randomness in decay, measurements will need to be taken in long time intervals and average value calculated for each set, B1
average may be different from true value as it is almost impossible to deduce when a nuclide will decay and what it will decay to. B1
which makes the measurements inaccurate

.....[2]

- (b) A stationary radon nucleus may undergo alpha decay to form the daughter nuclide X.

The masses of the nuclei and of the alpha particle are given in the Fig. 6.1.

nucleus or particle	mass / u
radon	222.0176
X	218.0090
alpha particle	4.0026

Fig. 6.1

- (i) Complete the nuclear decay equation, including all the decay products.



- (ii) Calculate the total kinetic energy of the products.

Total ke = (final mass – initial mass) x c ²	M1
= 9.0x10 ⁻¹³ J.	A1

total kinetic energy = J [2]

- (iii) Alpha particles can be stopped by tissue paper. Suggest if this implies that alpha-emitters present no health hazards in a school laboratory.

Alpha particles are highly ionising. Hence, if accidentally ingested or inhaled into the body, it can cause internal damage to cells, tissue etc. Therefore, although it has a short range, it can still be hazardous.

..... [1]

- (c) A theory of nuclear astrophysics proposes that all the elements heavier than iron are formed in supernova explosions ending the lives of massive stars. Assume equal amounts of ${}^{235}\text{U}$ and ${}^{238}\text{U}$ were created at the same time and the present $\frac{{}^{235}\text{U}}{{}^{238}\text{U}}$ ratio is 0.00725.

Calculate how long ago did the star explode that released the elements that formed our Earth.

Half lives of ${}^{235}\text{U}$ and ${}^{238}\text{U}$ are 0.704×10^9 years and 4.47×10^9 years respectively.

$\frac{e^{-\lambda_5 t}}{e^{-\lambda_8 t}} = 0.00725,$	M1
λ_5 – decay constant of ${}^{235}\text{U}$, λ_8 – decay constant of ${}^{238}\text{U}$	
Hence $t = 5.9 \times 10^9$ years	A1

time = years [2]

[Total: 10]

17

Section B

Answer **one** question from this Section in the spaces provided.

7. (a) There are quantitative and qualitative concepts of gravitational field that are analogous to those